

Emergency

State(FriendlyDeath&PavementEffectEmergency

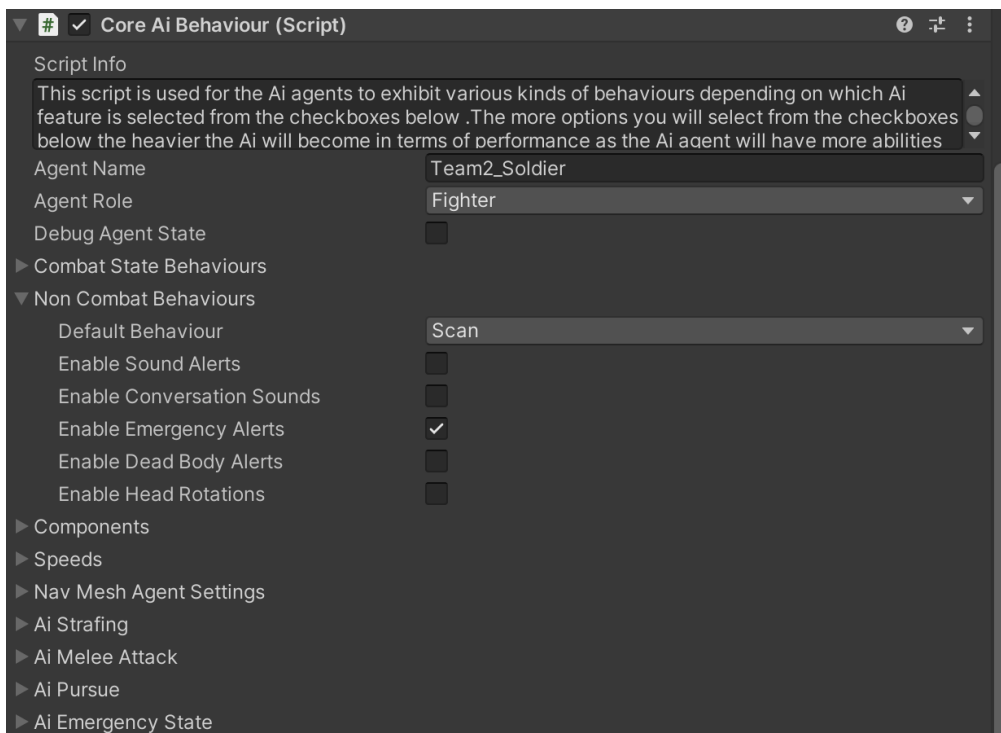
Part-6

Introduction

This is the Part-6 of AI Emergency State and in this part you will learn about how the AI get into emergency state when his friendly AI agent dies near him as well as when player misses the shot and spawn pavement impact effect around AI agent.

Enable Emergency Alert

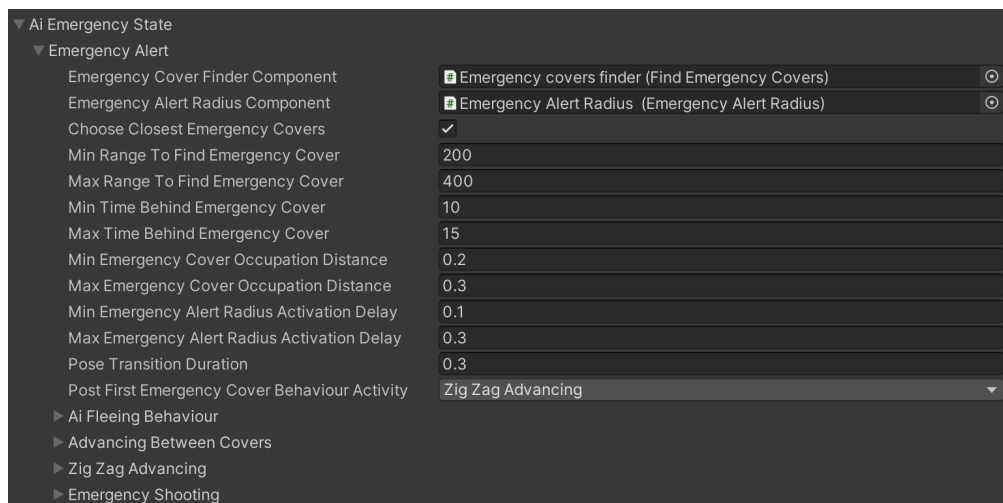
To enable emergency alert behaviour on AI agent you need to check the checkbox 'Enable Emergency Alerts' and enabling this checkbox will show the paragraph below called 'AI Emergency State'.



AI Emergency State

This paragraph is responsible for Ai agent behaviour in life threatening situations where he cannot detect the source of the threat and takes several measures to save his life

and locate the threat. Most common emergency state triggering situations are when Ai agent will hear bullet impacts near him or will hear dying sound of friendly Ai agent or when he will see another friendly Ai agent in emergency state. All those conditions will result in Emergency behaviour that Ai agent will execute in a various forms depending on two major conditions. When Ai agent gets into Emergency state he will be provided with the coordinate of the supposed enemy position. After that Navmesh path is calculated towards this coordinate. If navmesh path is complete then Ai agent will hide behind available emergency covers for specified time and when this hiding time is expired then Ai agent will advance between emergency covers towards supposed enemy position. If there are no more available emergency covers left for Ai agent on his way to supposed enemy position then Ai agent will start run in a zigzag pattern towards those coordinates to increase survival chances. If navmesh path is incomplete towards supposed enemy position (i.e if supposed enemy coordinate is on the roof of a different building or on the other side of the river etc.) then Ai agent will perform Emergency hiding and shooting in a general direction of the supposed enemy position. This behaviour can be set up that way so that Ai agent could switch emergency covers or stay at the first occupied emergency cover without ever leaving it.



Fields	Info
Emergency Cover Finder Component	Drag and drop Emergency covers finder game object with 'Find Emergency Covers' script attached to it from this Ai agent's hierarchy.
Emergency Alert Radius Component	Drag and drop Emergency covers finder game object with 'Emergency Alert Radius' script attached to it from this Ai agent's hierarchy.

Choose Closest Emergency Covers	In case this checkbox is active then Ai agent will find closest emergency covers during emergency state.
Min Range To Find Emergency Cover	Specifies the minimum emergency covers detection radius.
Max Range To Find Emergency Cover	Specifies the maximum emergency covers detection radius.
Min Time Behind Emergency Cover	Minimum time to hide behind emergency cover before beginning but not during advancing towards supposed enemy position.
Max Time Behind Emergency Cover	Maximum time to hide behind emergency cover before beginning but not during advancing towards supposed enemy position.
Min Emergency Cover Occupation Distance	Minimal Emergency Cover occupation distance reaching which Ai agent will stop moving towards it and start hiding behaviour.
Max Emergency Cover Occupation Distance	Maximum Emergency Cover occupation distance reaching which Ai agent will stop moving towards it and start hiding behaviour.
Min Emergency Alert Radius Activation Delay	Minimum amount in seconds of Ai agent's Emergency Alert Radius Trigger Activation delay.
Max Emergency Alert Radius Activation Delay	Maximum amount in seconds of Ai agent's Emergency Alert Radius Trigger Activation delay.
Pose Transition Duration	The duration, in seconds, it takes for the AI agent to rotate and focus on emergency cover or shooting points during emergency state. A lower value indicates a faster rotation, while a higher value results in a slower rotation.

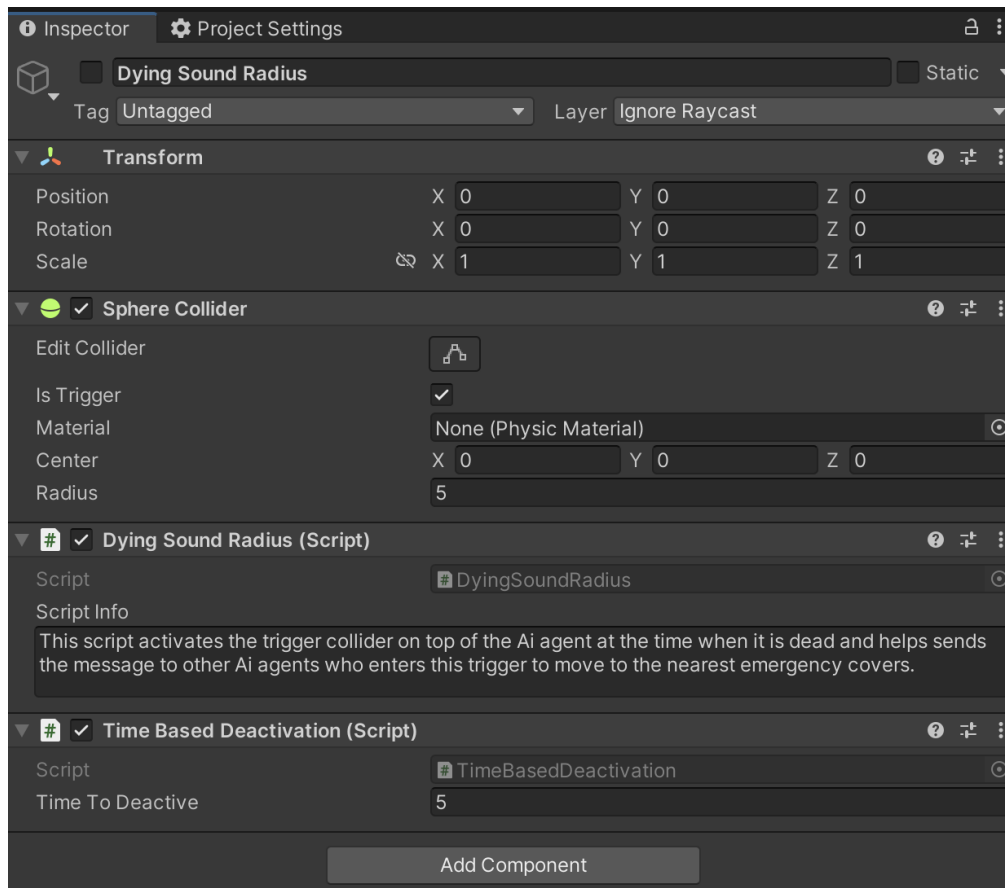
Post First Emergency Cover Behaviour Activity	Select any of the options from this drop down list that contains various behaviours for Ai agent to perform after expiration of the initial Emergency cover occupation timer.
Ai Fleeing Behaviour	This Subsection regulates the parameters of fleeing behaviour when Ai agent sprints away from danger zone when find himself in the emergency situation.
Advancing Between Covers	Fields of this subsection are regulating various aspects of Ai agent's advancing between covers towards enemy supposed coordinates.
ZigZag Advancing	This subsection sets the coordinate for the final destination of zigzag sprinting and regulates the amplitude of the zigzag sprinting pattern.
Emergency Shooting	This subsection regulates various aspects of Emergency Shooting behaviour which if activated will make Ai agent to fire his weapon in general direction of the enemy situated outside of enemy detection parameters of this Ai agent.

AI Dying Sound Radius

This script activates the trigger collider on top of the Ai agent at the time when it is dead and helps sends the message to other Ai agents who enters this trigger to move to the nearest emergency covers.

First create an empty gameObject to be the child of the Soldier Mesh and rename it to be 'Dying Sound Radius' and within the gameObject add a sphere Collider and two scripts to it called 'Dying Sound Radius' and 'Time Based Deactivation'. [See the Image Below]

- ▼  Team2_Soldier_1
 - ▶  metarig
 - ▼  Soldier_Mesh
-  Dying Sound Radius
- ▶  Retreat Directions
- ▶  Alert Radius
- ▶  Custom Strafe point
- ▶  Custom headlook points
- ▶  Custom Scan Aim Points
- ▶  Scan Aim Levels (During Non combat)
- ▶  Sounds
- ▶  Sound Alert Radius
- ▶  Weapon Sounds
-  Coordinate Creator
-  Covers finder
-  Firing Points Finder
-  Emergency covers finder
-  Patrol Points Finder
-  Conversation Sounds
-  Ai Voices
-  Roof Check Raycaster
-  Direct Enemy Raycaster
-  Horizontal Raycaster
-  Grenade Visibility Checker



Adding this script will allow the Dying AI agent to notify his friendly AI agent to get into the emergency state and using another script called 'Time Based Deactivation' script it will get deactivated after certain period of time using the value provided in the field 'Time to Deactive'.

Lastly, Make sure to drag and drop the gameObject in the field called 'Dying Sound Radius' exist in the 'Humanoid AI Health Script'. [See the image below]

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Humanoid Ai Health (Script)

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Script

HumanoidAiHealth

Script Info

This script is used for the Ai agent Health and death . It is responsible for spawning pickup items after Ai dies .

Health

40

Weak Point Damage Multiplier

5

Enable Health Bar

☒

Health Bar Prefab

Ai HealthBar

Stand Health Bar UI Offset

X 0

Y 1.8

Z 0

Crouch Health Bar UI Offset

X 0

Y 1.3

Z 0

Enable Ragdoll

☒

Animator Component

Team2_Soldier_1 (Animator)

▶ Stand Dying Animation State Names

3

▶ Crouch Dying Animation State Names

1

Head Bone Collider

head (Sphere Collider)

▶ Stand Head Shot Dying Animation State Names

1

▶ Crouch Head Shot Dying Animation State Names

1

▶ Hit Effects

1

Keep Dead Body

☒

Dead Body Mesh

Soldier_Mesh (Skinned Mesh Renderer)

▶ Dead Body Mesh Items To Attach

5

Pelvis Bone Transform

hips (Transform)

TIME_TO_DESTROY_OTHER_COMPONENT

4

Dead Body Existence Time

10

TIME_TO_UNPARENT_BODYMESH

5

Spawn Pickup Items After Death

☒

▶ Pickup Items

1

Reset Pickup Rotation

☒

Dying Sound Radius

Dying Sound Radius

Sound Investigating Approach Style Reset

None (Game Object)